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Featured Article

The Income Volatility of U.S. Commercial Farm Households

Nigel Key*, Daniel L. Prager, and Christopher B. Burns

The authors are economists at the USDA Economic Research Service. The views expressed are those of the authors and should not be attributed to the Economic Research Service or USDA.

*Correspondence to be sent to: nkey@ers.usda.gov

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Abstract *This study uses a newly created panel dataset drawn from the 1997 to 2013 Agricultural Resource Management Survey to provide the first national estimates of income volatility for commercial farm households in the United States. Results show that the income of commercial farm households is substantially more volatile than that of all U.S. households – though the volatility of farm income is not more volatile than income from nonfarm self-employment. Using a regression analysis, we identify operator, operation, and regional characteristics associated with higher income volatility, providing information that could improve targeting of risk-mitigating programs. We find that farm income volatility has declined for farms specializing in program crops in recent decades, supporting the hypothesis that the expansion of the federal crop insurance program helped reduce farm income risk.*

Key words: Income volatility, income variation, farm income, off-farm income, risk.

JEL codes: I31, J31, Q12.

Farm income is highly variable due to fluctuations over time in both yields and prices. For the 1.4 million U.S. farmers who consider farming their primary occupation, variability of returns can be a challenging part of running their business and providing for their families. Indeed, large, unplanned income fluctuations can affect the ability of farmers to obtain credit, expand their operations, and repay debt. Farmers may try to cope with income variation by borrowing or liquidating their assets; they may adjust their off-farm labor supply in anticipation of, or to compensate for, unexpected income shocks. Farmers might also cope with risk by adjusting their inputs or altering their crop mix. Hence, the way farmers cope with income risk affects what, how, and how much they produce, and can have important implications for agricultural production, rural household welfare, and environmental quality.

Federal agricultural policies have long sought to shelter farmers from income fluctuations using price supports, direct income support, disaster assistance programs, and crop yield and revenue insurance programs. Trade and agricultural policies that are mainly framed as supporting producer incomes and internalizing externalities can have important risk-reducing benefits for farmers (Thompson et al. 2004). Even the largely decoupled production flexibility contract (PFC) payments introduced by the 1996 FAIR Act provided insurance value to farmers by offering a relatively stable source of income. The 2014 Farm Act significantly shifted federal agricultural spending towards programs aimed at reducing income risk (USDA Economic Research Service 2016). The act ended fixed annual payments to producers based on historical production, and created new programs tied to annual or multi-year fluctuations in prices, yields, or revenues. The new programs include those that pay producers when prices fall below a reference price or revenue level (Price Loss Coverage (PLC) and Agriculture Risk Coverage (ARC), respectively), and expanded crop insurance programs aimed at providing support for shallow revenue or yield losses.

Despite the important role of income volatility in determining farm household behavior and welfare, and despite the growing emphasis of federal programs on income risk reduction, there exists little empirical information about the extent of U.S. farm household income volatility, how this volatility varies across different types of households, or the role of government programs in mitigating farm household income volatility. This article has two main objectives. The first is to measure the income volatility of households that operate commercial farms in the United States. Key questions include how much does income vary from year to year after households have mitigated income variation through off-farm work, futures markets, contracts, etc.? And how does the income variability faced by farm households compare to that of all U.S. households? This information can help inform policymakers seeking to understand farm households' need for risk-mitigating programs.

The second objective of the paper is to identify how income volatility varies across different types of farm households. Information about how operator characteristics such as age, educational attainment, and marital status influence volatility could be useful in targeting information about risk management programs. Information about how volatility is correlated with farm commodity specialization and local economic conditions could help identify the types of producers who could benefit most from new risk mitigating programs or expanded insurance availability.

The dearth of information about farm household income variability is largely attributable to a lack of data tracking farm household income over time—that is, farm household panel data. Past studies of farm income variability at the national level have relied on either aggregate or cross-sectional data. Aggregate data can provide useful insight into how the sector as a whole fares from year to year, but can mask considerable variation at the farm level (Mishra and Sandretto 2002). In a given year, producers in one region might be thriving, whereas those in another region might be incurring losses from local drought or pest infestations.

Studies using cross-sectional data (e.g., Mishra and El-Osta 2001; Mishra et al. 2002) also only provide limited information about individual farm income variability. Inter-annual variation in commodity prices, policies, and yields can sometimes result in "boom" or "bust" cycles, causing the average income of farmers to change from year to year. Examining variation in income across farms at one point in time ignores this inter-annual variation, and therefore underestimates individual farm income variation.

The drawbacks associated with aggregate and cross-sectional data can be addressed with farm-level panel data spanning several years. We construct such a panel by matching observations of farms that were surveyed more than once between 1997 and 2013 by the USDA Agricultural Resource Management Survey (ARMS) —the most comprehensive survey of U.S. farm households. The panel nature of the data allows us to observe how farm and nonfarm income and program payments changed over time for the same household, which allows for an accurate assessment of inter-annual income fluctuations. Because the ARMS was not designed as a panel, the sample of repeat observations used in this study are not representative of the farm population as a whole. However, as we show, the farms we observe display characteristics that are very similar, on average, to commercial farms. Hence, the study provides insight into income volatility for the types of farms responsible for most agricultural production in the United States.

While the existing information about farm household income volatility is limited, there have been a number of studies that have used panel data to examine the income volatility of all U.S. households—usually seeking to identify how volatility varies across income categories and over time. Early studies focused on decomposing the cross-sectional variance in individual earnings into permanent and transitory components, and on identifying time trends using the Panel Study of Income Dynamics (Gottschalk and Moffitt 1994; Haider 2001) or the Current Population Survey (Cameron and Tracy 1998). More recent studies have examined trends in household income volatility using simple measures of volatility, which are usually a function of the change in income over consecutive years (e.g., Congressional Budget Office 2008; Dahl, DeLeire, and Schwabish 2011; Moffitt and Gottschalk 2011; Shin and Solon 2011; Ziliak, Hardy, and Bollinger 2011; Dynan, Elmendorf, and Sichel 2012).

In this study we adapt some of the measures of volatility used in the recent studies of all (mostly nonfarm) households to allow for the negative farm and total incomes experienced by some farm households. The measures allow us to compare the volatility of farm and off-farm income sources, and to compare the total income volatility of farm to all households. We use a regression analysis to determine how different farm and operator characteristics influence farm and total income volatility, and we investigate trends in volatility over time.

Data

The farm-level panel dataset is constructed with data from the 1997–2013 Agricultural Resource Management Surveys (ARMS). The ARMS is an annual USDA survey carried out by the National Agricultural Statistics Service (NASS) and Economic Research Service (ERS) (USDA-ERS 2015a). Although the ARMS is not a panel survey, some farmers were surveyed multiple times over the eighteen year span due to either random chance or

Table 1 Number of Times the Same Farm Is Observed in ARMS between 1996 and 2013

Number of Times Observed	Distinct Farms	Percentage of Distinct Farms Observed
1	190,732	83.3
2	29,511	12.9
3	6,648	2.9
4	1,705	0.7
5	396	0.2
6	66	<0.1
7	13	<0.1
8	2	<0.1
Total	229,073	100.0

Source: Agricultural Resource Management Survey (ARMS), 1996-2013.

their agricultural importance within their state. We identify these repeat observations using the NASS operator identification number.[1]

Of a total of 229,073 farmers surveyed between 1996 and 2013, 37,945 were surveyed more than once. Of these repeat observations, 29,511 were surveyed twice, 6,648 three times, and 1,786 were surveyed four or more times (table 1). Our approach is to compare changes in real income across two periods.[2] For farmers who were sampled more than twice, each interior pair of years was used to create an observation.[3] We drop observations where the span between the observations is greater than 5 years in order to keep the time between observations relatively homogeneous, and omit 1996 due to insufficient observations. We also drop observations if the difference in operator age between two observations was more than 7 years (which would imply that a different person is operating the farm).[4] We limit the study to family farms—operations where the operator and the operator's family own the majority of the business. These family farms represent about 98% of all farms.

Finally, because the farm households that were surveyed at least twice between 1997 and 2013 tend to operate much larger farms and to produce more output than an average farm, we limit the sample to farms categorized by ERS as "commercial farms" in either year they were surveyed. According to the ERS farm typology in place for most of the study period, commercial farms include family farms with gross sales of at least $250,000 per year

[1]*The NASS uses a multipart method to track operations and operators over time. For family farm operations without hired professional managers, the principal operator is tracked over time. For "managed operations" that use hired farm managers, the operation is tracked and household information, including information on principal operator household income, is not collected in most years. Managed operations and non-family farms are not included in this study.*

[2]*All values are deflated to 2011 U.S. dollars using the Bureau of Economic Analysis' Gross Domestic Product Implicit Price Deflator, available online at: http://research.stlouisfed.org/fred2/series/GDPDEF/*

[3]*For example, a farmer surveyed in 1998, 2003, and 2006 would be included twice in the panel data set: once for the period between 1998 and 2003, and once for the period between 2003 and 2006.*

[4]*In most cases, the operator identification number is updated when there is a change in the principal operator. However, in some cases the operator identification number is not updated despite a change in the person making day-to-day decisions on the farm. This can occur in situations where the operation of the farm passes from one generation to another (e.g., from the father to the son).*

(Hoppe and MacDonald 2013). The final panel sample consists of 20,319 observations. Table 2 displays the distribution of the final sample across the two years that farms were observed.

All statistics are calculated using ARMS sampling weights that account for the probability of selection within a given year—giving a weight to an individual farm that is inversely proportional to their probability of selection and adjusting for nonresponse. These sample weights will also account for the probability of selection across the different years in our panel, placing a higher weight on observations from years with a smaller sample size.[5]

To account for the original ARMS sample design we bootstrap the standard errors presented in the summary statistics and regression results. The ARMS sample is stratified by sales class (and commodity) within individual estimate states or regions. We create 150 replicate weights that are stratified based on the sales class of the operation using the "bsweights" package developed by Kolenikov (2010). In our regressions, we also include fixed effects for both state and commodity type.

Table 3 displays some key household and farm characteristics for: (*a*) the panel sample, (*b*) all farms that were surveyed between 1997 and 2013, and (*c*) the subset of the full ARMS sample that are categorized as "commercial farms" according to the ERS farm typology (Hoppe and MacDonald 2013). As compared with the full ARMS sample (column 2), the panel sample of commercial farms (column 1) and the full sample of commercial farms (column 3) tend to operate much larger farms and produce more. The commercial farms received less income from off-farm sources and significantly more from farming. These farms also had an average household income that was more than twice that of the average farm. The panel sample has characteristics that are very similar to the full sample of commercial farms. The fact that the farms in the panel are comparable to commercial farms implies that our analysis should provide insight into the income volatility for the larger-scale operations that produce almost 80% of agricultural output.[6]

Measuring Income Volatility

To calculate farm, off-farm, and total household income using the ARMS data we employ the same methods used by the USDA ERS (USDA ERS 2015b). Farm income is defined as the sum of the operator household's share of farm business income (net cash farm income less depreciation), wages paid to the operator and other household members, and net rental income from renting farmland. In addition, some households report other farm-related income from operating a farm business other than the one being surveyed, and in-kind payments to household members for farm work.

Off-farm income comes from earned and unearned sources. Earned income includes compensation from wages and salaries for household members and net earnings from operating nonfarm businesses. Unearned income is derived from interest and dividends from investments, transfer payments

[5]*Because the population size of farms is relatively consistent throughout this period, the average sample weights in a year when 9,000 farms were sampled will be approximately twice that in a year when 18,000 farms were sampled.*

[6]*In 2010, commercial farms represented 9.9% of all farms and produced 79.0% of total output (Hoppe and MacDonald 2013).*

Table 2 Distribution of Final Sample Observations across Years

						Year 2								
Year 1	**2000**	**2001**	**2002**	**2003**	**2004**	**2005**	**2006**	**2007**	**2008**	**2009**	**2010**	**2011**	**2012**	**2013**	**Total**
1997	–	136	163												299
1998	199	78	112	140											529
1999	73	174	173	212	153										785
2000		41	278	253	207	257									1,036
2001			21	259	172	196	168								816
2002				74	376	333	287	223							1,293
2003					40	682	495	435	392						2,044
2004						162	820	408	469	382					2,241
2005							308	858	624	413	424				2,627
2006								143	875	596	421	539			2,574
2007									131	748	492	460	374		2,205
2008										89	802	557	500	353	2,301
2009											99	896	472	–	1,467
2010												102	–	–	102
Total	272	429	747	938	948	1,630	2,078	2,067	2,491	2,228	2,238	2,554	1,346	353	20,319

Source: Agricultural Resource Management Survey (ARMS), 1997–2013.
Note: This table shows the distribution of observations used in the analyses. The sample excludes pairs of observations collected more than five years apart.

Table 3 Summary Statistics for Panel, All ARMS Farms, and Commercial Farms, 1997–2013

	Panel sample Mean (Std. Dev.)	**All ARMS farms Mean (Std. Dev.)**	**ARMS commercial farms Mean (Std. Dev.)**
Farm income	140,140	11,864	153,210
	(377,487)	(171,714)	(549,144)
Off-farm income	49,971	76,199	53,239
	(101,249)	(132,791)	(167,232)
Total household income	190,111	88,064	206,449
	(393,860)	(215,621)	(579,980)
Farm assets	2,475,864	735,509	2,402,701
	(3,680,085)	(1,707,252)	(4,256,969)
Total assets	2,797,893	889,594	2,278,741
	(3,826,528)	(1,737,893)	(4,120,865)
Total debt	477,820	115,781	445,397
	(837,325)	(338,627)	(887,684)
Value of production	1,107,944	112,326	989,288
	(1,993,505)	(673,554)	(2,080,434)
Crop farm (1/0)	0.555	0.337	0.584
	(0.497)	(0.473)	(0.493)
Operator age	51.6	56.7	52.6
	(10.9)	(13.4)	(11.6)
Observations	20,319	261,146	105,464

Source: Agricultural Resource Management Survey (ARMS), 1997–2013.
Note: The "panel sample" consists of farms surveyed more than once between 1997 and 2013. "All ARMS farms" include all farms surveyed between 1997 and 2013. "All ARMS commercial farms" include all ARMS farms that are categorized "commercial" according to the ERS typology (Hoppe and MacDonald 2013). All values are deflated to 2011 dollars. Values for the panel sample are average values for the two years surveyed. All averages account for yearly sampling weights.

such as pensions and unemployment benefits from both private and public sources, and other off-farm income, such as gifts or bequests.

As enumerated later, farm income and total household income can be negative for some households in some years. Some commonly used measures of volatility, such as the coefficient of variation or the percentage change in income become nonsensical when income (or average income) is negative. In this section, we describe several measures of income volatility that allow for negative income values.

One such measure is the absolute value of the income change between periods, $|y_{it} - y_{is}|$, where y_{is} and y_{it} are the incomes of household i in years s and t. Another easily interpreted measure of income dispersion is the standard deviation of income: $\sqrt{\left(y_{is} - \bar{y}_i\right)^2 + \left(y_{it} - \bar{y}_i\right)^2}$, where $\bar{y}_i$ is average income in the years s and t. While these two measures describe the magnitude of income fluctuations over time, they do not take into account the size of the change relative to expected income. It is likely that a given income shock would have different welfare and behavioral implications depending on a household's expected income. For this reason, it is common to scale measures of income change by average income.

The absolute value of the arc percent change (AAPC) is one such income-scaled measure:

$$AAPC_i = 100 * \left| \frac{y_{it} - y_{is}}{\bar{y}_i} \right|. \tag{1}$$

The arc percentage change is often preferred to the percentage change as a measure of income volatility because the arc percentage change is symmetric regarding increases or decreases in income and it is bounded between -200 and 200 (Dyan, Elmendorf, and Sichel 2012; Hardy and Ziliak 2014). The second factor is particularly important when dealing with a skewed income distribution that includes large changes from year to year. The AAPC is bounded by 0 and 200.

The coefficient of variation (CV) is a second measure of income volatility that is scaled by average income. If the CV is large, then income varies widely relative to the mean, whereas if it is small then income usually falls within a narrow range around the mean. The absolute value of the coefficient of variation (ACV) of income allows for possible negative mean income values:

$$ACV_i = \left| \frac{\sqrt{\left(y_{is} - \bar{y}_i\right)^2 + \left(y_{it} - \bar{y}_i\right)^2}}{\bar{y}_i} \right|. \tag{2}$$

Unlike the AAPC, the ACV is not bounded. When households have a very small average income, the ACV can be extremely large, which can skew regression parameters.[7] To address this problem, we use the natural logarithm of the ACV as the dependent variable in the regressions. The log transformation reduces the influence of the outliers and makes data conform more closely to the normal distribution.

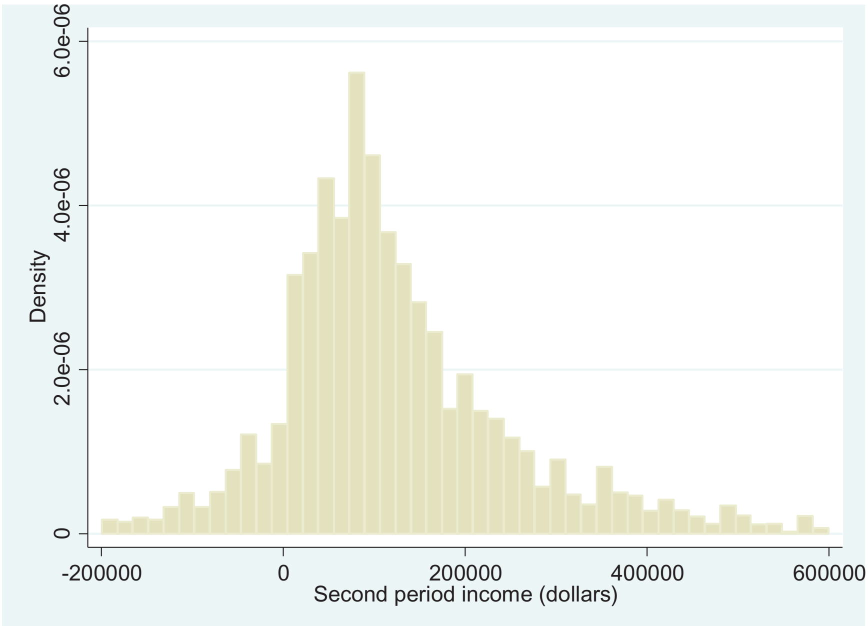
A measure that is commonly used to examine trends in volatility for the broad population of households is the standard deviation across households of the arc percentage change (SDAPC) (Dahl, DeLeire, and Schwabish 2011; Ziliak, Hardy, and Bollinger 2011; Dyan, Elmendorf and Sichel 2012). Unlike the previous measures, the SDAPC does not measure volatility for an individual household, but rather for the sample, or a subsample, at one point in time:

$$SDAPC_t = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(APC_{it} - \overline{APC}_t\right)^2}, \tag{3}$$

where $APC_{it} = \frac{(y_{it} - y_{is})}{0.5 \cdot (|y_{it}| + |y_{is}|)}$ and N is the number of households in the sample.

[7]*For instance, consider a small farm household that earns $20,000 one year and suffers a loss of $18,000 the next year. This corresponds with an average income of $1,000 but a standard deviation of 26,870. The ACV is 26.9, which is unusually large.*

Figure 1 Second period total household income for households with first period total income between $75,000 and $125,000
Source: Agricultural Resource Management Survey (ARMS), 1997–2013.
Note: Positive and negative outliers are truncated in the figure for clarity.



Results

Income Volatility

To illustrate the scale of household income volatility for a typical commercial farm household, consider the distribution of second-period income for households that earned between $75,000 and $125,000 in the first period (figure 1).[8] The figure shows that the average second-period income centers on the average value of first-year income (about $100,000), but varies substantially. In fact, most households earn less than $75,000 or more than $125,000—the initial range of income in the first year. A significant share of households have net incomes less than $0 or more than $200,000 in the second period.

The high volatility of total household income illustrated in figure 1 is driven mainly by farm rather than off-farm income. For the same group of middle-income households (those that earned a total income between $75,000 and $125,000 in the first year), net farm income varies widely in the second period and a substantial share of households experienced negative net farm income (figure 2). In contrast, almost all second-period off-farm income was positive, and the distribution of off-farm income was clustered between $0 and $100,000 (figure 3).

The measures of income volatility defined in the previous section confirm that farm income is much more volatile than off-farm income (table 4, column 1). For the full panel sample, the median absolute change between periods for farm income is $117,693, which is 61% more than the median farm income ($73,144). In contrast, the median absolute change in off-farm

[8]*A total of 2,971 households representing 14.6% of the sample earned income in this range.*

Figure 2 Second period farm income for households with first period total income between $75,000 and $125,000
Source: Agricultural Resource Management Survey (ARMS), 1997–2013.
Note: Positive and negative outliers are truncated in the figure for clarity

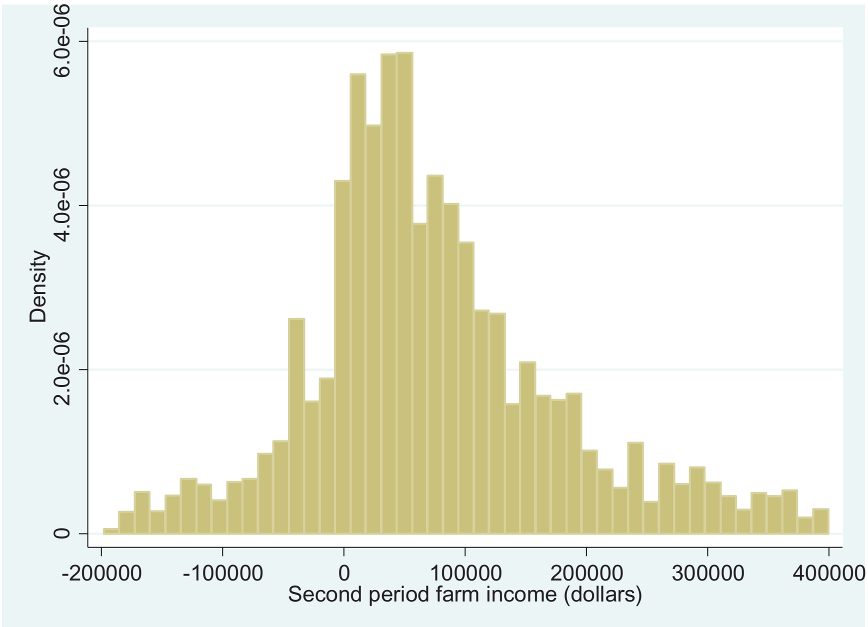
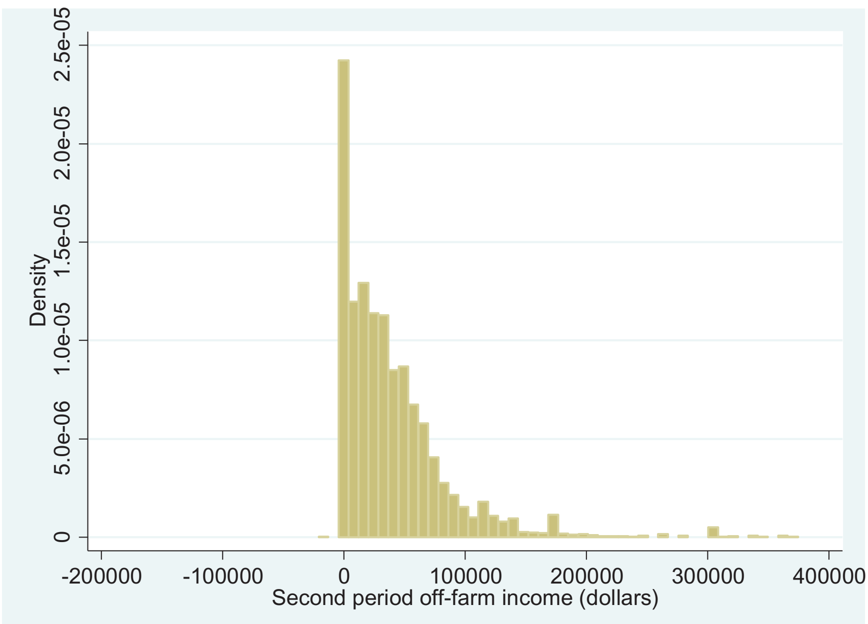


Figure 3 Second period off-farm income for households with first period total income between $75,000 and $125,000
Source: Agricultural Resource Management Survey (ARMS), 1997–2013
Note: Positive and negative outliers are truncated in the figure for clarity.



income is $15,185, which is about half of median off-farm income ($29,729). While 40% of households in the sample experienced negative farm income in at least one of the two periods, less than one tenth of one percent of the

Table 4 Volatility Measures of Crop, Livestock and All Farms, 1997–2013

	Full sample (Span = 1-5)	95% Confidence Interval for Selected Variables	Predicted value (Span = 1, Midyear = 2005)	Span = 2
Farm Income				
Median	73,144		73,458	69,032
Median absolute change between years	117,693		113,258	111,143
Mean	140,140	[133,848, 146,432]	146,707	142,239
Mean absolute change between years	264,937	[256,625, 273,249]	270,878	274,113
Share negative in at least one year	0.40		.42	0.40
Share negative in both years	0.08		.09	0.08
Mean absolute arc percent change	126.9	[125.5, 128.3]	125.4	126.6
Mean absolute coefficient of variation	1.37	[1.32, 1.41]	1.37	1.34
Std. dev. arc percentage change	143.5		n.a.	143.1
Off-farm Income				
Median	29,729		28,723	30,892
Median absolute change between years	15,185		13,974	15,496
Mean	49,971	[48,071, 51,871]	48,151	50,081
Mean absolute change between years	43,740	[41,647, 45,833]	39,999	43,409
Share negative in at least one year	0.00		0.00	0.00
Share negative in both years	0.00		0.00	0.00
Mean absolute arc percent change	95.9	[94.2, 97.6]	95.4	95.0
Mean absolute coefficient of variation	0.68	[0.67, 0.69]	0.68	0.71
Std. dev. arc percentage change	124.9		n.a.	124.2
Total Household Income				
Median	115,702		115,777	116,554
Median absolute change between years	127,052		119,421	121,397
Mean	190,111	[183,337, 196,885]	194,858	192,320
Mean absolute change between years	279,228	[270,683, 287,774]	283,625	287,786
Share negative in at least one year	0.29		0.29	0.30
Share negative in both years	0.04		0.04	0.04
Mean absolute arc percent change	110.4	[108.8, 112.0]	110.1	110.3

Continued

Table 4 *Continued*

	Full sample (Span = 1-5)	95% Confidence Interval for Selected Variables	Predicted value (Span = 1, Midyear = 2005)	Span = 2
Mean absolute coefficient of variation	1.11	[1.08, 1.15]	1.12	1.16
Std. dev. arc percentage change	131.5		n.a.	131.9
Observations	20,319		20,319	6,967

Source: Agricultural Resource Management Survey (ARMS), 1997–2013. Averages account for sampling weights.
Note: The standard deviation of the arc percentage change is not observed for individual households so is not predicted by the econometric model (column 3). Confidence intervals are created using bootstrapped standard errors.

sample had negative off-farm income in either period. Reflecting the greater variation in income relative to the mean, the AAPC in farm income averaged 127, compared to 96 for off-farm income.[9] Similarly, the ACV of farm income is 1.34 versus 0.72 for off-farm income.

As discussed in the data section, the sample is comprised of pairs of observations surveyed between one and five years apart. It is possible that the measures of income variability change with the time span between observations – presumably a longer span would cause a somewhat larger change in income, and therefore greater measured volatility. Almost all studies of U.S. household income volatility used data collected in consecutive years – that is, with a span equal to one. To better compare our estimates of farm household income volatility to estimates from other studies we adjust our estimates using the following model:

$$y_i = \alpha + \theta Span_i + \gamma Year_i + \epsilon_i, \tag{4}$$

where y_i is a measure of income or income volatility for household i, $Span_i$ is the number of years between surveys for that household, and $Year_i$ is the midpoint year between surveys. Including the span in the regression allows us to predict volatility when the span is one. Including the midpoint year allows us to control for possible volatility trends over time. Since the ARMS survey has more observations in the later years compared to the earlier years (as shown in table 2), if volatility were trending up or down, then the unequal number of surveys by year could influence the estimates.

Column 3 in table 4 presents the predicted values of various income and income volatility measures when the span = 1 and midpoint year = 2005. The predicted values in column 3 are very similar to the unadjusted values in column 1, suggesting that controlling for span and time trend has only a minimal effect on the estimates. We use the results in column 3 where possible to compare with the estimates for all households.

[9]*The absolute arc percent change can be interpreted as the absolute value of the arc percentaage change (which is similar to the percentage change). So a value of 127 means the absolute value of the change from year to year was about 125% larger than the mean value in the two years.*

Another way to demonstrate that the volatility measures are robust to span length is to compute the measures using only one span. Column 4 shows the estimates calculated for the subset of farms that are observed two years apart—the most common gap between observations—comprising 34.2% of the sample. The span = 2 estimates are very similar to the unadjusted and adjusted estimates for the full sample shown in columns 1 and 3.

Farm Households versus All U.S. Households

Our estimates show that commercial farm households have much more volatile income than the typical U.S. household. Using data from the Current Population Survey, Hertz (2006) reports that the median absolute change in U.S. household income between consecutive years was $11,345 in 2003–04, which was approximately 25% of median income at the time. In contrast, the farms in the ARMS panel had a predicted median absolute income change between consecutive years of $119,421, which was 103% of median income (table 4, column 3).

Farm households are also more likely to experience very large income changes than are typical U.S. households. Dahl, DeLeire, and Schwabish (2011) find that about 9% of all U.S. households had income changes of at least 50% between consecutive years. In contrast, we find that among the 85% of households with positive household income in the first year, two-thirds (66%) had a total household income change of at least 50%.[10]

Furthermore, Dynan, Elmendorf, and Sichel (2012) find the standard deviation of the arc percent change of household income averaged about 50% since the mid-1990s using data from the Panel Study of Income Dynamics. Dahl, DeLeire, and Schwabish (2011) find the same measure averaged around 30% using the Survey of Income and Program Participation and Social Security Administration data. These values are substantially below the 131% that we estimate (table 4, column 1), confirming that farm households have much more volatile income than typical households.

While the total household income of commercial farmers is more volatile than for the average U.S. household, the volatility of farm income does not appear to be more volatile than income from nonfarm self-employment. A Congressional Budget Office study (2008) found that the standard deviation of the arc percentage change in self-employment income ranged between 140 and 150 from 1992 to 2003, which is similar in magnitude to the variability of farm income (143).

The higher total income volatility of farm households is driven partly by the fact that farmers receive most of their income from on-farm sources—and, as we discussed above, this income is more volatile than off-farm income. However, there appears to be more to the story: the nonfarm income of farm households is also relatively volatile. For farm households, the standard deviation of the arc percent change of off-farm income is 125%—well above the total income volatility of all households (which is 30% to 50%). Similarly, among the 86% of farm households that reported positive off-farm income, 56% had off-farm income that changed by at least 50% (49% of those surveyed in consecutive periods also had large income changes). This is much higher

[10] *Only farms with positive income in the first period are considered because the percent change is not defined if first period income is negative or zero. This statistic is for all farms. For farms that were surveyed in consecutive years, we find that 58 percent had their income change by at least 50 percent.*

Table 5 Income Volatility by Farm and Operator Characteristics

	Total income		**Farm income**		**Off-farm income**	
	Mean	**ACV**	**Mean**	**ACV**	**Mean**	**ACV**
Farm asset category						
< $900K	129,275	1.04	82,771	1.39	46,503	0.66
$900K – $1.6M	146,412	1.09	102,065	1.33	44,347	0.66
$1.6M – $3.0M	187,763	1.15	142,067	1.38	45,696	0.68
> $3.0M	347,128	1.23	278,765	1.35	68,363	0.73
Highest education attained (1/0)						
Less than high school diploma	140,019	1.14	103,838	1.58	36,181	0.79
High school	168,515	1.13	128,142	1.34	40,372	0.70
Some college	180,733	1.09	134,880	1.34	45,853	0.66
Four or more years of college	239,472	1.11	169,474	1.39	69,999	0.65
Operator married (period 1)						
No	146,282	1.22	113,269	1.60	33,013	0.88
Yes	195,881	1.10	143,760	1.34	52,121	0.66
Farm Type						
Switched commodities	158,547	1.25	112,289	1.43	46,259	0.70
Cash grains	186,758	1.07	139,210	1.34	47,548	0.63
Rice	162,144	0.98	121,635	1.20	40,508	0.81
Cotton	291,248	1.03	225,533	1.12	65,715	0.68
High value	330,598	1.16	264,100	1.31	66,498	0.79
Other crops	203,165	1.15	157,088	1.30	46,077	0.72
Hogs	182,769	1.11	137,208	1.28	45,561	0.61
Poultry	95,864	0.93	47,537	1.52	48,326	0.62
Dairy	210,814	1.13	178,963	1.28	31,851	0.80
Cattle and general livestock	176,274	1.31	100,899	1.52	75,375	0.64

Source: Agricultural Resource Management Survey (ARMS), 1997–2013.
Note: ACV = Absolute Coefficient of Variation. Means and ACV account for sampling weights. Farm type is assigned based on a contribution of 50% or more to the total value of production.

than the 9% of all households who had their income change by at least 50% (Dahl, DeLeire, and Schwabish 2011). It is possible that farm households have particularly volatile off-farm income because their off-farm labor decisions are influenced by their highly volatile farm income. That is, farm households may make frequent adjustments to their off-farm employment to compensate for the large fluctuations in their farm income (Mishra and Goodwin 1997).

Factors Associated with Income Volatility

Table 5 shows how income and income volatility varies across different types of farm households. In terms of farm size (measured by farm assets) there is a clear positive correlation between farm size and off-farm income and total income volatility. However, there is no obvious relationship between farm size and farm income volatility. Income volatility of all types is lower for better-educated operators. However, this effect diminishes after an operator has some college. Married operators have substantially less off-farm, farm, and total income volatility than non-married operators.

The table also shows income volatility by commodity specialization, which is assigned if a commodity comprises 50% or more to the total value of production in both survey years. Farms that did not specialize in the same commodity category in both years are assigned to the "switched commodities" category. Poultry and cattle farms have the highest farm income volatility, while cotton farmers have the lowest, on average. Hog farmers have the lowest off-farm income volatility while dairy and rice farmers have the highest. In terms of total income volatility, poultry farmers have the lowest, while cattle farms have the highest.

The summary statistics provide information about income volatility for the average farm household with a particular characteristic. However, the statistics do not allow us to isolate the effect of particular farm or operator characteristics on volatility. For example, poultry farms have higher than average farm income volatility. However, poultry farms also have less-educated operators on average, and less well-educated operators have higher average farm income volatility. It is not clear whether the high farm income volatility for poultry farmers is due to the commodity specialization or other factors such as their operator education. To identify how farm and operator characteristics influence farm, off-farm, and total income variability we use a regression of the form:

$$Volatility_i = \alpha + \theta Span_i + \gamma Year_i + \boldsymbol{\beta X_i} + \delta \mathrm{State_i} + \epsilon_\mathrm{i}, \quad (5)$$

where $\boldsymbol{X_i}$ includes exogenous grower and operation characteristics and measures of local economic conditions, and $State_i$ is a state dummy variable that is included to account for soil quality, climate, and other time-invariant regional effects. The parameter γ on the mid-year variable indicates the annual rate of change in volatility.

A few earlier studies have used regression analyses to explore the volatility of farm business income using data from certain U.S. States, Canada, or Europe (e.g., Schurle and Tholstrup 1989; Purdy, Langemeier, and Featherstone 1997; Poon and Weersink 2011; Enjolras et al. 2014). A possible critique of these studies is that they included some regressors that are endogenously determined with income volatility. Potentially endogenous variables include measures of risk mitigation such as government program participation, farm enterprise diversification, borrowing, and off-farm labor participation. These variables are not only likely to affect income risk, but also to be influenced by income risk. For example, farmers who face higher production and income risks (e.g., from pest or weather hazards) are more likely to purchase crop insurance, diversify their production, borrow more, or work more off-farm. As a result, we might observe a positive correlation between income risk and these risk mitigation strategies in a regression, even though the strategies lower risk compared to what it would be otherwise. Therefore, it is impossible to meaningfully interpret the estimated parameters or determine the direction of causality when endogenous variables are included in the regression. For this reason, we include only plausibly exogenous variables that are likely to be determined independently of income risk.

Table 6 presents summary statistics for the variables used in the regression. As shown in the table, the average span between surveys was about 3 years. We use 10 commodity specialization categories, with those that switched specialization between survey years as the missing category. The majority of farms (61%) specialized in crop production. The sample is close to evenly divided among the four farm size (asset) categories. Most

Table 6 Summary of Regression Variables

Variable	Mean	Std. Dev.
Income volatility (Log of Abs. Coeff. of Var.)		
Total	−0.373	1.425
Farm	−0.096	1.422
Off-farm	−0.892	1.251
Mid-year between surveys	2005.5	3.228
Span between surveys	3.111	1.202
Farm Type (categorical)		
Switched commodities	0.119	0.323
Cash grains	0.289	0.453
Rice	0.014	0.119
Cotton	0.028	0.165
High value	0.090	0.287
Other crops	0.068	0.235
Hogs	0.050	0.217
Poultry	0.130	0.337
Dairy	0.123	0.328
Cattle and general livestock	0.089	0.285
Farm asset category		
< \$900K	0.303	0.460
\$900K – \$1.6M	0.259	0.438
\$1.6M – \$3.0M	0.245	0.430
> \$3.0M	0.193	0.394
Highest education attained (1/0)		
Less than high school diploma	0.062	0.242
High school	0.386	0.487
Some college	0.268	0.443
Four or more years of college	0.283	0.451
Operator age	51.56	10.90
Operator Married (period 1)	0.890	0.313
Population Interaction Index Change (2010–2000)	2280	3969
Observations	20,319	

Source: Agricultural Resource Management Survey (ARMS), 1997–2013. Means account for sampling weights.
Note: Farm type is assigned based on a contribution of 50% or more to the total value of production.

operators (94%) completed high school and 28% completed at least four years of college. Operators have an average age of 51.6. About 89% of the operators were married the first year they were surveyed.

We use a population trend measure to characterize local economic conditions.[11] Population trends within the county are measured using the difference in the Population Interaction Index (PII) between 2010 and 2000. The PII is derived from a gravitational model of population density that provides a continuous measure of proximity to nearby population centers (Ribaudo and Johansson 2007).

We use the natural logarithm of the absolute coefficient of variation (ACV) as our main measure of volatility. Using the ACV as the dependent variable produces similar results in terms of parameter significance and

[11] *A local area unemployment rate was also tried, but was not found to be statistically significant, so was dropped from the regression.*

sign, but the logarithm of the ACV permits interpreting the coefficients in terms of percentage change and better fits the data.[12] As a robustness check, we also estimate the model using the natural logarithm of absolute arc percentage change (AAPC) as the dependent variable. Because the AAPC is censored above 200 and below 0, we estimate a Tobit regression model. As a second robustness check, we estimate the model using the subset of observations which are observed two years apart (i.e., span = 2). The alternative specifications produce very similar results to the base specification.[13]

Researchers using ARMS normally account for the sample design in estimating variances using a jackknife method with replicate weights provided by the USDA/NASS. For the panel data this is an unattractive option because the replicate weights are designed for the cross-sectional sample. For this reason, we create sampling weights that are based on the average sampling weight assigned to the farm in each of the two years that it was sampled. We then use the "bsweights" package (Kolenikov 2010) to create 150 boostrap replicate weights, stratified based on the sales class of the operation. For each of our regressions, we use boostrapped standard errors based on the replicate weights.

Regression Results

Holding the other exogenous factors constant, the regression results indicate that specializing in certain commodities for both survey years is associated with significantly less volatile farm income than switching commodity specialization (the missing category; table 7). For example, specializing in rice, cotton, and high value crops was associated with at least 20% less volatile farm income.[14] High value crops (mainly fruits and vegetables) might display lower farm income variability because these crops are often produced under a marketing contract arrangement in which prices are negotiated before harvest, which lowers price risks for the farmer.

Specializing in poultry and dairy production was also associated with lower farm income variability. Poultry is produced almost exclusively under production contracts, which generally eliminates most price risk for the grower. Dairy producers are sheltered from some farm income risk by several federal programs that were functioning during most of the survey period. For example, under the Milk Income Loss Contract Program authorized by the 2002 Farm Bill, payments were made to dairy operations when milk prices fell below a certain level. In addition, the Dairy Product Price Support Program specified support prices for milk, and later (following the 2008 Farm Bill) manufactured milk-based products. Prices were supported by government purchases of these products.

In terms of farm size, larger farms (i.e., farms with more assets) generally earn more farm income and experience larger absolute changes in farm income that do smaller farms. However, as the regression shows, in proportion to average farm income, farm income volatility does not differ

[12]The ACV is not bounded and there are a number of outliers. Taking the natural log of the ACV reduces the influence of outliers and makes the data more normal in shape, thus satisfying the assumptions of an OLS regression.

[13]Results of the robustness checks are available from the authors upon request.

[14]Because the volatility measures are in logarithmic form, the percentage change in the volatility measures given a one unit change in the independent variable is calculated as $100 \cdot (\exp(\beta) - 1)$, where β is the estimated coefficient.

Table 7 Regression Analysis: What Factors Explain Income Variation?

VARIABLES	(1) Total Income	(2) Farm Income	(3) Off-farm Income
Mid-year	−0.0187***	−0.0156***	−0.0180***
	(0.00595)	(0.00546)	(0.00527)
Year span	0.00726	0.00836	0.0284**
Commodity specialization:	(0.0146)	(0.0151)	(0.0134)
Cash Grains	−0.0520	−0.0770	−0.146***
	(0.0555)	(0.0538)	(0.0564)
Rice	−0.244*	−0.304**	−0.0218
	(0.129)	(0.129)	(0.124)
Cotton	−0.177*	−0.438***	−0.130
	(0.104)	(0.0889)	(0.0981)
High-value crops	−0.149**	−0.222***	0.174**
	(0.0639)	(0.0645)	(0.0677)
Other crops	−0.0509	−0.160**	−0.0149
	(0.0736)	(0.0734)	(0.0680)
Hogs	0.0203	−0.0315	−0.228**
	(0.0865)	(0.102)	(0.0897)
Poultry	−0.305***	−0.165**	−0.219***
	(0.0713)	(0.0658)	(0.0624)
Dairy	−0.174***	−0.323***	0.168***
	(0.0658)	(0.0644)	(0.0610)
Other livestock	−7.37e-05	−0.112	−0.164**
Farm assets:	(0.0905)	(0.0809)	(0.0804)
$900,000 - $1.6m	0.0730	−0.0283	0.0262
	(0.0516)	(0.0451)	(0.0449)
$1.6m - $3.0m	0.167***	0.0532	0.108**
	(0.0522)	(0.0499)	(0.0463)
More than $3.0m	0.203***	−0.0136	0.224***
Operator education:	(0.0523)	(0.0461)	(0.0444)
High school	−0.142	−0.0719	−0.239***
	(0.103)	(0.0925)	(0.0770)
Some college	−0.200**	−0.0946	−0.323***
	(0.0996)	(0.0943)	(0.0811)
College or more	−0.239**	−0.0929	−0.392***
Operator age in first year:	(0.102)	(0.0953)	(0.0773)
50-65	−0.0467	−0.0533	0.0254
	(0.0358)	(0.0373)	(0.0349)
>65	0.0282	0.135***	−0.213***
	(0.0508)	(0.0472)	(0.0460)
Operator Married: Year 1	−0.292***	−0.194***	−0.507***
	(0.0537)	(0.0575)	(0.0398)
Population Interaction Change	−7.56e-06**	1.48e-06	−2.67e-06
	(3.85e-06)	(4.37e-06)	(3.85e-06)
Constant	37.88***	31.69***	35.82***
	(11.96)	(10.96)	(10.57)
Observations	20,265	20,243	19,058
R-squared	0.027	0.023	
State FEs	Yes	Yes	Yes

Source: Authors' calculations using data from the Agricultural Resource Management Survey (ARMS).
Note: State-clustered standard errors appear in parentheses. Asterisks indicate the following:
$*** = p<0.01$, $** = p<0.05$, and $* = p<0.1$).

significantly across farm sizes. In contrast, the riskiness of off-farm income does increase substantially with farm size: the largest farms (those with at least $2.6 million in assets) have off-farm income volatility that is 25% greater than small farms. It is possible that larger farms, with more assets and higher average incomes, are able to indulge in riskier off-farm investments. Alternatively, these households with large farm businesses are more likely to use the off-farm labor market as a response to farm income shocks rather than as a constant source of income.

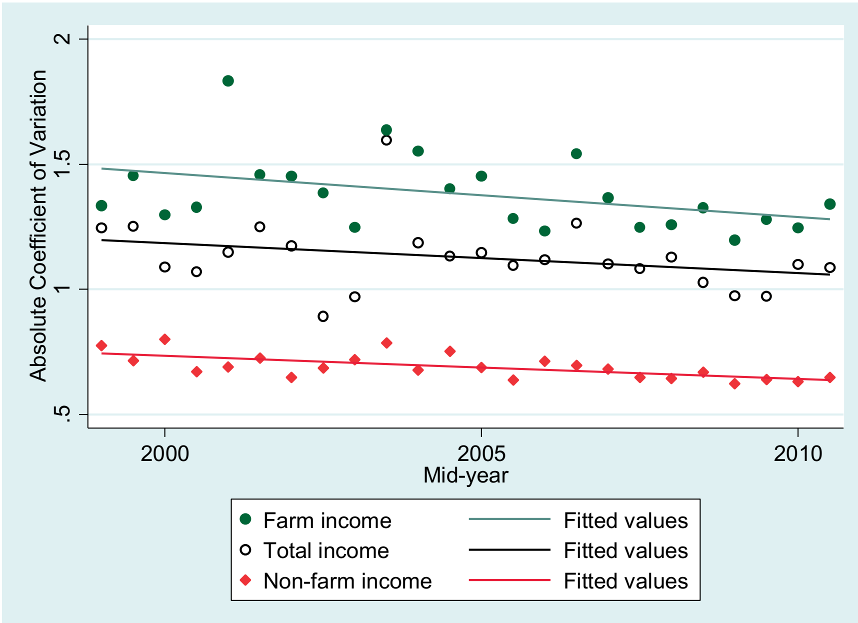
The income sources of farm households differ from all households in ways that can explain the difference in total income volatility. Households operating large farms have a greater share of their total income coming from farm income, which is riskier than off-farm income, and they have more volatile off-farm income. These two factors combine to make total household income more volatile for larger farms compared to smaller farms: households with at least $3.0 million in farm assets have total income volatility that is 22% greater than the smallest farms. The higher volatility means that households operating larger farms are more likely to experience years with negative income despite having higher average income. The probability of having negative household income in at least one of the two observed periods is 17% for the smallest farms compared to 33% for the largest farms.

The finding that larger (and higher average income) farms have more volatile household income contrasts with (mostly) nonfarm households, where studies have consistently found less income volatility among higher-wage earners and higher-income households (Hertz 2006; Dahl, DeLeire, and Schwabish 2011; Moffitt and Gottschalk 2011; Shin and Solon 2011; Hardy and Ziliak 2014).

Results also indicate that off-farm income and total income volatility are substantially lower if the principal operator has more education. Operators with a college degree had total household income volatility that was 27% lower and off-farm income volatility that was 39% lower than those who did not graduate from high school. Operators with some college education had total income that was 22% lower and off-farm income that was 38% lower. Compared to having only a high school diploma, having four years of college reduced total income volatility by 10% and off-farm income volatility by 16%. Higher income volatility for less educated farmers is consistent with findings for all households (Ziliak, Hardy, and Bollinger 2011; Dynan, Elmendorf, and Sichel 2012). Individuals with less education may have more volatile income because they have less employment security. This study covers a period that spanned the Great Recession—during which less-educated workers faced larger increases in unemployment (and hence, negative off-farm income shocks) than better-educated workers (Hout and Cumberworth 2012).

Operators older than 65 had farm income volatility that was 14% higher than operators younger than 50, holding other factors constant. On the other hand, those over 65 had off-farm income volatility that was 23% lower than those under 50; their off-farm income might be less volatile because older farmers are more likely to be retired from off-farm occupations, and to qualify for stable retirement annuities and Social Security payments. Older farmers' farm income might be more volatile because of large negative changes in farm income as they transition out of farm work and into retirement. Alternatively, farm income risk might be higher for older farmers because their more stable off-farm income allows them to take more risks on-farm.

Figure 4 Trends in the absolute coefficient of variation of farm, off-farm and total income
Source: Agricultural Resource Management Survey (ARMS), 1997–2013.
Note: Each data point represents the average income volatility for each survey midpoint in the sample. The survey midpoint is defined as (year 2 – year 1)/2.



Marital status had a large effect on income volatility. Being married was associated with a 66% decrease in off-farm income volatility and a 34% decrease in total household income volatility. Compared to single individuals, married couples likely have a larger share of household labor earning income from less volatile off-farm sources, which reduces total income variability.

The local economic conditions, as measured by the population interaction index, had a significant influence on total household income volatility. Living in an area with an increasing/decreasing population density was associated with lower/higher levels of total income volatility. The finding that farmers who reside in counties with declining populations have more volatile income suggests that programs aimed at supporting the rural economy may also be effective at decreasing income volatility for farm households—in addition to traditional farm policies aimed at reducing farm income risk.

Trends in Income Volatility

The negative and significant coefficients on the mid-year variable (the midpoint between the two years the farm was surveyed) indicates that the volatility of farm income declined about 1.6% per year, while total income and off-farm income declined by 1.8% per year. These regression results, which control for changes in the composition of the panel sample over time are also evident in a plot of farm, off-farm and total income volatility (ACV) over the span of the dataset (figure 4). Because each observation is a single measure of income volatility between two years, we use the midpoint of those two years as the date on the graphs. Hence, each point represents the average ACV for all the farms at that midpoint. The figure also shows an estimated linear trend for each income component. The graph shows a declining trend in volatility for all income types.

Studies of the volatility trends in the United States find more income variability in the 1980s than in the 1970s, and flat trends in variability during the 1980s and early 1990s—though these studies differ in their findings in more recent periods (Congressional Budget Office 2008; Dahl, DeLeire, and Schwabish 2011; Moffitt and Gottschalk 2011; Shin and Solon 2011; Ziliak, Hardy, and Bollinger 2011; Dynan, Elmendorf, and Sichel 2012). However, because farms are influenced by different economic conditions and policies than nonfarm households, it is plausible that farm households would have experienced different trends in income volatility.

It is possible that farm policies have been increasingly effective in reducing the income volatility of farmers over the study period because of increasing resources being directed towards the federal crop insurance program. Over this period the number of acres enrolled beyond the most basic catastrophic coverage level increased from 117 million acres in 1996 to 280 million acres in 2013 and federal subsidies to purchase insurance increased from $720 million to $6.6 billion in constant 2009 dollars over the same period (RMA 2015).

To test the hypothesis that farm policies have contributed to the decline in income volatility, we create a new category of farms that specialize in program crops—the crops that receive the bulk of farm payments and federal crop insurance subsidies. Farms in the “program crops” include those specializing in “cash grains,” rice, and cotton. This category does not include farm specializing in any of the livestock commodities, high value crops (fruits and vegetables), or “other crops.” We interact the program crop indicator with the time (mid-year) variable to see if volatility changed over time at a different rate for this group. Results show that program crop producers saw their farm income volatility decline about 2% per year more rapidly than other types of farms—which saw no statistically significant decline in volatility over time (table 8). For off-farm income (and total income) the negative trends in volatility were not associated with specializing in program crops.

The finding that program crop producers have lower farm income volatility but not lower off-farm volatility is consistent with the hypothesis that changes in federal farm programs contributed to lower farm income volatility over time. However, there are other possible factors that could have contributed this trend. For example, it is possible that changes in farming technologies and practices used on program crops (e.g., increased use of genetically engineered crops, no-till farming, GPS technologies, precision agriculture, irrigation, etc.) reduced yield variation over time.

Young Beginning Farmers

Finally, we consider the effect of farming experience on income volatility. It is plausible that farm income volatility is higher for new farmers, and that over time farmers learn techniques that allow them to mitigate their income risks. To test whether beginning farmers have more variable income, we include an indicator for whether a farmer has no more than 10 consecutive years of farming experience (the USDA definition of a beginning farmer) in the main regression model. We find that the coefficient on the indicator

Table 8 Regression with Program Crop Indicator Interacted with Time Trend

VARIABLES	(1) Total Income	(2) Farm Income	(3) Off-farm Income
Mid-year	−0.0160**	−0.00850	−0.0173***
	(0.00655)	(0.00678)	(0.00633)
Year span	0.00756	0.00912	0.0285**
	(0.0146)	(0.0151)	(0.0135)
Program crops * Mid-year	**-0.00798**	**-0.0209***	**-0.00190**
Commodity specialization:	**(0.0137)**	**(0.0122)**	**(0.0111)**
Cash Grains	15.96	41.91*	3.674
	(27.45)	(24.57)	(22.25)
Rice	15.76	41.66*	3.797
	(27.46)	(24.57)	(22.24)
Cotton	15.83	41.53*	3.688
	(27.42)	(24.56)	(22.25)
High-value crops	−0.150**	−0.225***	0.174**
	(0.0638)	(0.0647)	(0.0678)
Other crops	−0.0509	−0.160**	−0.0148
	(0.0736)	(0.0735)	(0.0679)
Hogs	0.0193	−0.0341	−0.228**
	(0.0863)	(0.102)	(0.0896)
Poultry	−0.305***	−0.165**	−0.219***
	(0.0714)	(0.0661)	(0.0624)
Dairy	−0.173***	−0.322***	0.168***
	(0.0659)	(0.0646)	(0.0609)
Other livestock	0.000156	−0.111	−0.164**
Farm assets:	(0.0905)	(0.0811)	(0.0803)
$900,000 - $1.6m	0.0721	−0.0307	0.0260
	(0.0514)	(0.0451)	(0.0450)
$1.6m - $3.0m	0.167***	0.0524	0.108**
	(0.0520)	(0.0499)	(0.0465)
More than $3.0m	0.202***	−0.0145	0.224***
Operator education:	(0.0521)	(0.0461)	(0.0444)
High school	−0.142	−0.0716	−0.239***
	(0.102)	(0.0916)	(0.0770)
Some college	−0.200**	−0.0937	−0.323***
	(0.0992)	(0.0936)	(0.0812)
College or more	−0.239**	−0.0924	−0.391***
Operator age in first year:	(0.101)	(0.0945)	(0.0773)
50-65	−0.0472	−0.0546	0.0253
	(0.0356)	(0.0372)	(0.0350)
>65	0.0280	0.135***	−0.213***
	(0.0508)	(0.0472)	(0.0460)
Operator Married: Year 1	−0.293***	−0.197***	−0.507***
	(0.0536)	(0.0574)	(0.0401)
Population Interaction Change	−7.52e-06*	1.58e-06	−2.67e-06
	(3.84e-06)	(4.39e-06)	(3.84e-06)
Constant	32.44**	17.46	34.50***
	(13.18)	(13.58)	(12.71)
Observations	20,265	20,243	19,058
R-squared	0.027	0.023	
State Fes	Yes	Yes	Yes

Source: Authors' calculations using data from the Agricultural Resource Management Survey (ARMS).
Note: State-clustered standard errors appear in parentheses. Asterisks indicate the following:
*** = $p<0.01$, ** = $p<0.05$, and * = $p<0.1$).

variable is not statistically significantly different from zero, indicating that beginning farmers as a group do not have more volatile income than more experienced farmers, after controlling for age and other characteristics.[15]

However, beginning farmers are a diverse group. If we focus instead on young beginning farmers (those under age 50) we find a very different result. We find that being a young and beginning farmer is associated with a 36% increase in farm income volatility and a 20% increase in total income volatility after controlling for other operator and operation characteristics. We find no statistically significant difference for off-farm income volatility.

The USDA has a number of programs that target beginning farmers, including Farm Service Agency (FSA) "Beginning Farmer" direct and guaranteed loan programs, which are aimed, in part, at helping beginning farmers to start or expand their operations. The finding that young beginning farmers have particularly variable income could serve as additional justification for federal loan program support for beginning farmers.

Conclusion

This study used a newly-created panel dataset drawn from the 1997 to 2013 Agricultural Resource Management Survey to provide the first national estimates of income volatility for commercial farm households. Because each household in the sample is observed at least twice, the data allowed us to obtain a more accurate measure of household income fluctuations than would be possible using cross-sectional or aggregate income data.

The data show that commercial farm households have much more variable household income than do typical U.S. households. The median change in total income between consecutive years was about eight times larger for the farm households than the typical U.S. household. The household income for farmers is highly volatile mainly because farm income varies much more than off-farm income—though the variability of farm income is comparable to that of the self-employment income of non-farm households.

Recent decades have seen a shift in agricultural spending towards programs that reduce farm income risk. This shift includes increases in federal subsidies for crop insurance that resulted in increases in coverage and acres enrolled. In more recent years, crop insurance programs have expanded to non-traditional program crops and livestock. In addition, the 2014 Farm Act ended fixed annual payments to producers based on historical production, and created new programs tied to annual or multi-year fluctuations in prices, yields, or revenues. Policymakers seeking to further expand the farm safety net could benefit from information about the types of producers who are most vulnerable to household income risk.

This study identified some farm household characteristics associated with greater household income variability. In particular, we found that income volatility increases with farm size—unlike for typical U.S. household for which income volatility declines with average income. Total household income is more volatile on larger farms because operators of larger farms derive a greater share of household income from the farm, and because they have more volatile off-farm income. We also found that household income

[15]*The beginning farmer variable has been generated by ERS only since 2005, so the sample used in the regression only includes 8,702 observations. Results are not shown in table 8, but are available from authors upon request.*

is more volatile when the principal operator has less education, is unmarried, or is a young and beginning farmer.

Local economic conditions also appear to influence income volatility: a higher local population density growth rate decreases total income volatility. This suggests that, in addition to traditional risk mitigation farm policies, economic development programs aimed at depressed areas of the rural economy may be effective at reducing income volatility for farm households.

The study also shed light on recent trends in farm income volatility. Results indicate that the volatility of farm income declined about 1.6% per year, while total income and off-farm income declined by 1.8% per year between 1997 and 2013. We also find that those specializing in program crops (the commodities associated with the bulk of agricultural program payments) saw a significantly greater decline in volatility than those not specializing in program crops. These results are consistent with the hypothesis that the shift in federal spending toward risk-mitigating agricultural policies has lowered farm income variability. However, there are other possible factors, such as changes in farming technologies that could have contributed to this trend.

The results illustrate the magnitude of the temporal income fluctuations experienced by commercial-scale farm households. Agricultural programs play an important and valuable role in helping farmers cope with these fluctuations. While income volatility appears to have declined over the last two decades, the paper demonstrates that even with increased spending on risk-mitigating agricultural programs, most commercial farm households continue to face substantial income risk.

References

Cameron, S., and J. Tracy. 1998. *Earnings Variability in the United States: An Examination Using Matched-CPS Data*. Unpublished paper, Federal Reserve Bank of New York.

Congressional Budget Office. 2008. Recent Trends in the Variability of Individual Earnings and Household. Available at: http://www.cbo.gov/publication/41714.

Dahl, M., T. DeLeire, and J. Schwabish. 2011. Estimates of Year-to-Year Volatility in Earnings and in Household Incomes from Administrative, Survey, and Matched Data. *Journal of Human Resources* 46 (4): 750–74.

Dynan, K., D. Elmendorf, and D. Sichel. 2012. The Evolution of Household Income Volatility. *The B.E. Journal of Economic Analysis and Policy* 12 (2): 1–40.

Enjolras, G., F. Capitanio, M. Aubert, and F. Adinolfi. 2014. Direct Payments, Crop Insurance and the Volatility of Farm Income. Some Evidence in France and in Italy. *New Medit: A Mediterranean Journal of Economics, Agriculture and Environment* 13 (1): 31–40.

Gottschalk, P., and R. Moffitt. 1994. The Growth of Earnings Instability in the U.S. Labor Market. *Brookings Papers on Economic Activity* 2: 217–72.

Haider, S. 2001. Earnings Instability and Earnings Inequality of Males in the United States, 1967-1991. *Journal of Labor Economics* 19: 799–836.

Hardy, B., and J. Ziliak. 2014. Decomposing Trends in Income Volatility: The "Wild Ride" at the Top and Bottom. *Economic Inquiry* 52 (1): 459–76.

Hertz, T. 2006. *Understanding Mobility in America.* The Center for American Progress. Discussion Paper.

Hoppe, R., and J. MacDonald. 2013. *Updating the ERS Farm Typology*. Washington DC: U.S. Department of Agriculture, Economic Information Bulletin No. EIB-110.

Hout, M., and E. Cumberworth. 2012. *The Labor Force and the Great Recession.* Stanford, CA: The Russel Sage Foundation and the Stanford Center on Poverty and Inequality.

Kolenikov, S. 2010. Resampling Variance Estimation for Complex Survey Data. *Stata Journal* 10 (2): 165–99.

Mishra, A., and H. El-Osta. 2001. A Temporal Comparison of Sources of Variability in Farm Household Income. *Agricultural Finance Review* 61 (2): 181–98.

Mishra, A., and B. Goodwin. 1997. Farm Income Variability and the Supply of off-Farm Labor. *American Journal of Agricultural Economics* 79 (8): 880–7.

Mishra, A., and C. Sandretto. 2002. Stability of Farm Income and the Role of Nonfarm Income in US Agriculture. *Review of Agricultural Economics* 24 (1): 208–21.

Mishra, A., H. El-Osta, M. Morehart, J. Johnson, and J. Hopkins. 2002. *Income, Wealth, and the Economic Well-Being of Farm Households.* Washington DC: U.S. Department of Agriculture, Economic Research Service, Agricultural Economic Report No. 812.

Moffitt, R., and P. Gottschalk. 2011. *Trends in the Transitory Variance of Male Earnings in the US, 1970–2004*. Working papers, The Johns Hopkins University, Department of Economics, No. 578.

Poon, K., and A. Weersink. 2011. Factors Affecting Variability in Farm and Off-farm Income. *Agricultural Finance Review* 71 (3): 379–97.

Purdy, B., M. Langemeier, and A. Featherstone. 1997. Financial Performance, Risk, and Specialization. *Journal of Agricultural and Applied Economics* 27 (1): 149–61.

Ribaudo, M., and R. Johansson. 2007. Nutrient Management Use at the Rural-Urban Fringe: Does Demand for Environmental Quality Play a Role? *Review of Agricultural Economics* 29: 689–99.

Schurle, B., and M. Tholstrup. 1989. Farm Characteristics and Business Risk in Production Agriculture. *North Central Journal of Agricultural Economics* 11 (2): 183–8.

Shin, D., and G. Solon. 2011. Trends in Men's Earnings Volatility: What Does the Panel Study of Income Dynamics Show? *Journal of Public Economics* 95 (7): 973–82.

Thompson, S., P.M. Schmitz, N. Iwai, and B. Goodwin. 2004. The Real Rate of Protection: The Income and Insurance Effects of Agricultural Policy. *Applied Economics* 36 (16): 1851–8.

United States Department of Agriculture (USDA), Economic Research Service (ERS). 2015a. ARMS Farm Financial and Crop Production Practices. Available at: http://www.ers.usda.gov/data-products/arms-farm-financial-and-crop-production-practices.aspx (accessed May 11, 2015).

——. 2015b. Farm Household Well-being: Income and Wealth in Context. Available at: http://www.ers.usda.gov/topics/farm-economy/farm-household-well-being/income-and-wealth-in-context.aspx (accessed May 19, 2015).

——. 2016. Projected Spending under the 2014 Farm Bill. Available at: http://www.ers.usda.gov/topics/farm-economy/farm-commodity-policy/projected-spending-under-the-2014-farm-bill.aspx (accessed June 3, 2016).

U.S. Department of Agriculture, Risk Management Agency (RMA). 2015. Summary of Business. Available at: http://www.rma.usda.gov/data/sob.html (accessed December 1, 2015).

Ziliak, J., B. Hardy, and C. Bollinger. 2011. Earnings Volatility in America: Evidence from Matched CPS. *Labour Economics* 18 (6): 742–54.